

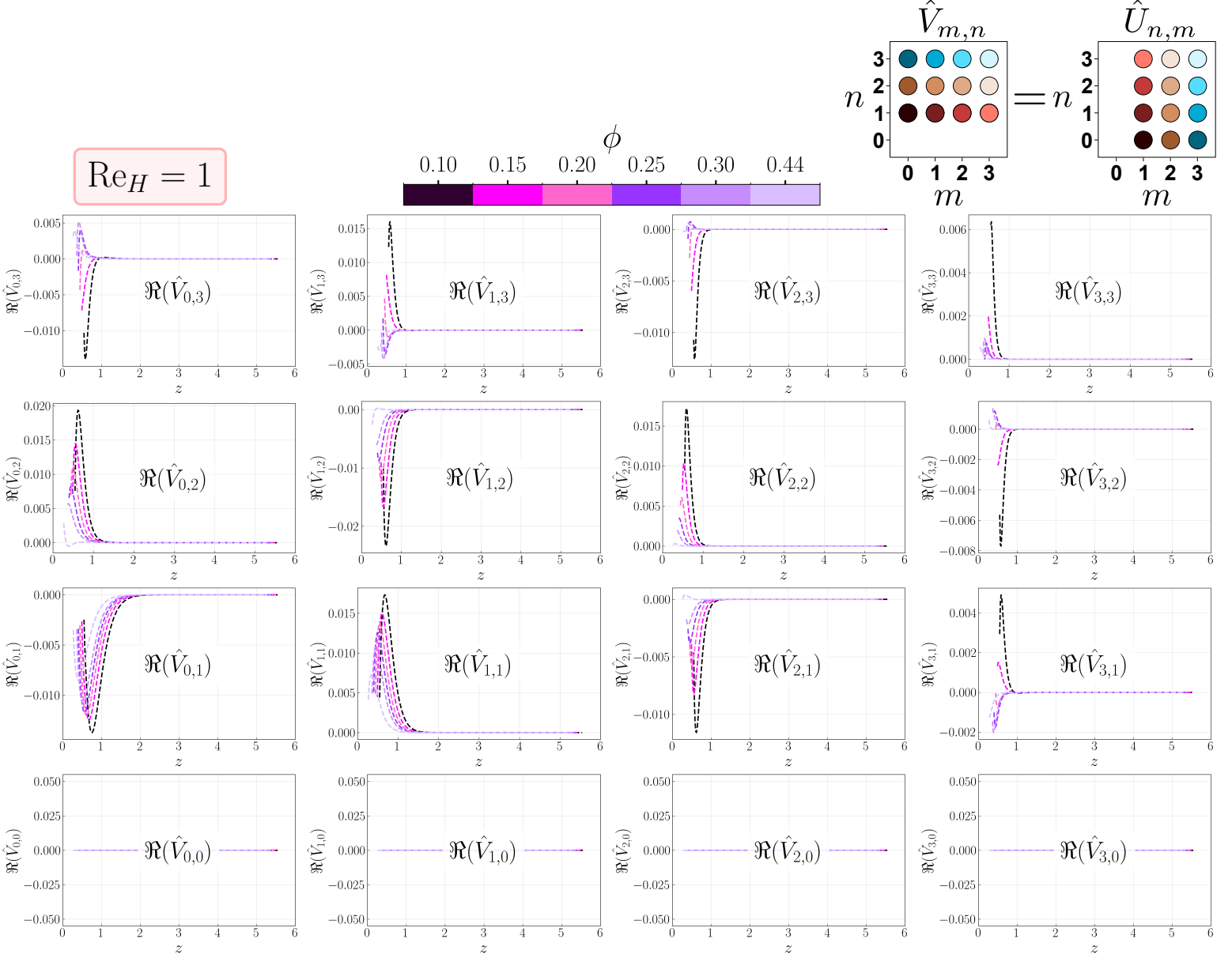


Figure B.27: Streamwise evolution of the real part $\mathcal{R}(\hat{V}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 1$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

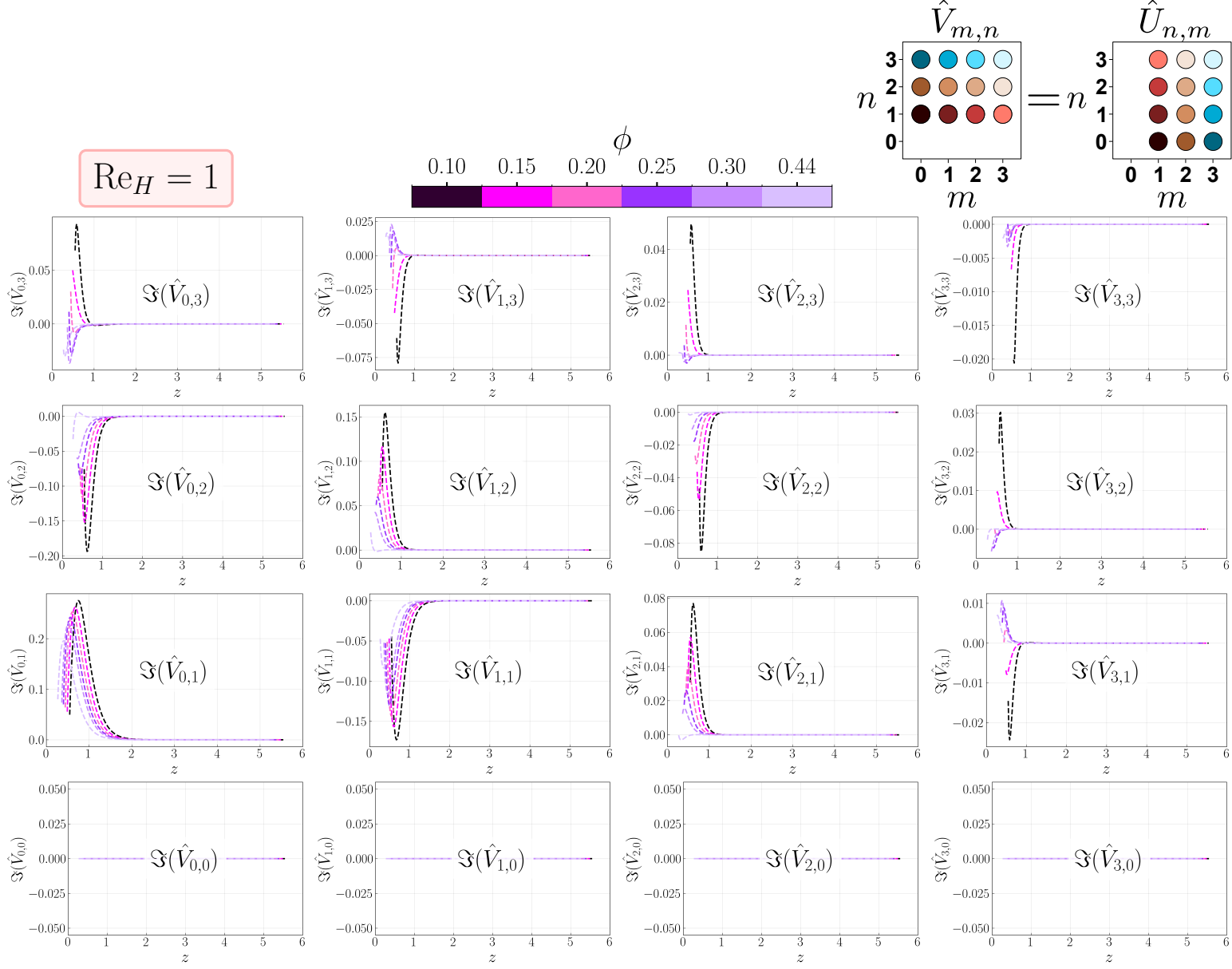


Figure B.28: Streamwise evolution of the imaginary part $\Im(\hat{V}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 1$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

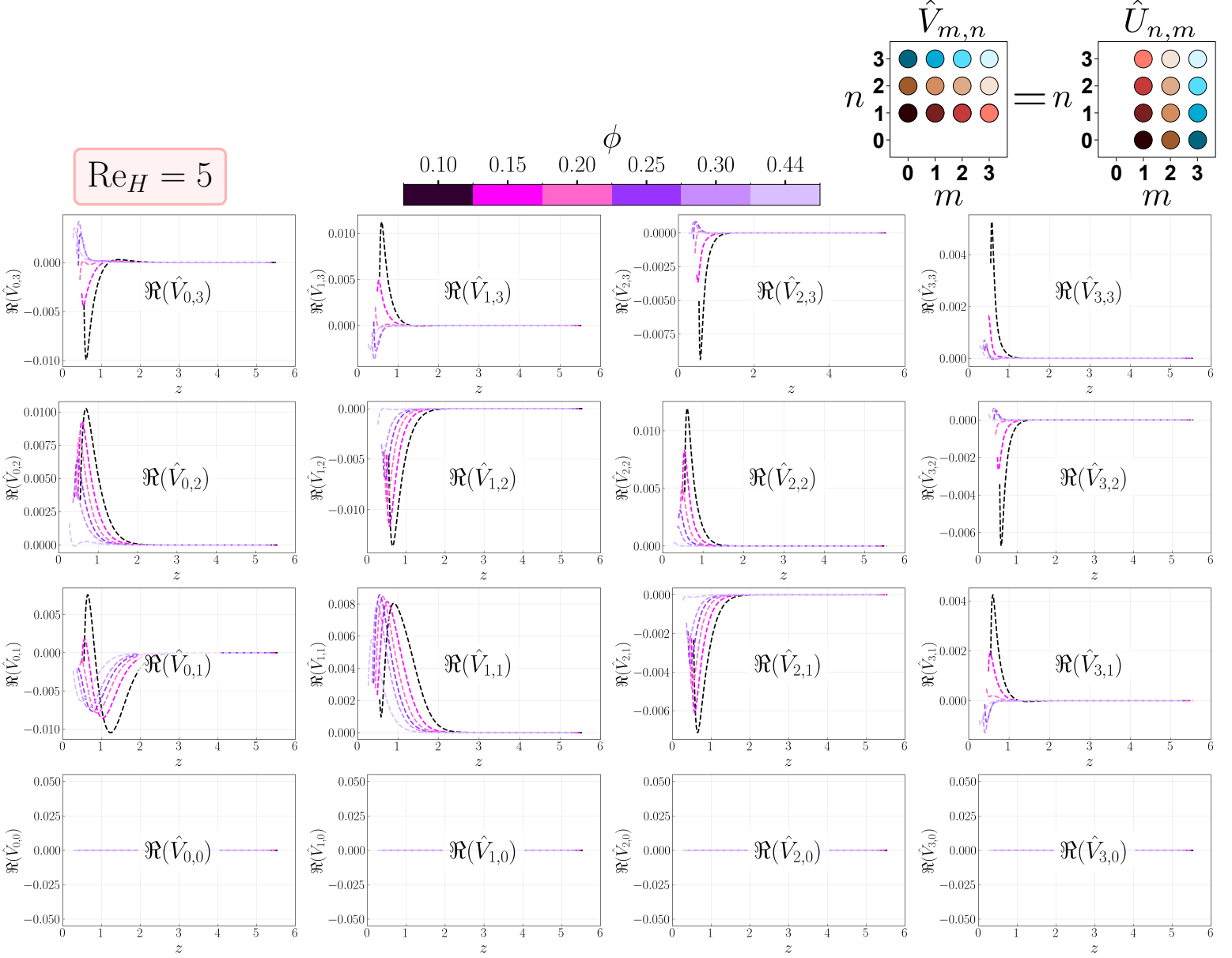


Figure B.29: Streamwise evolution of the real part $\Re(\hat{V}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 5$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

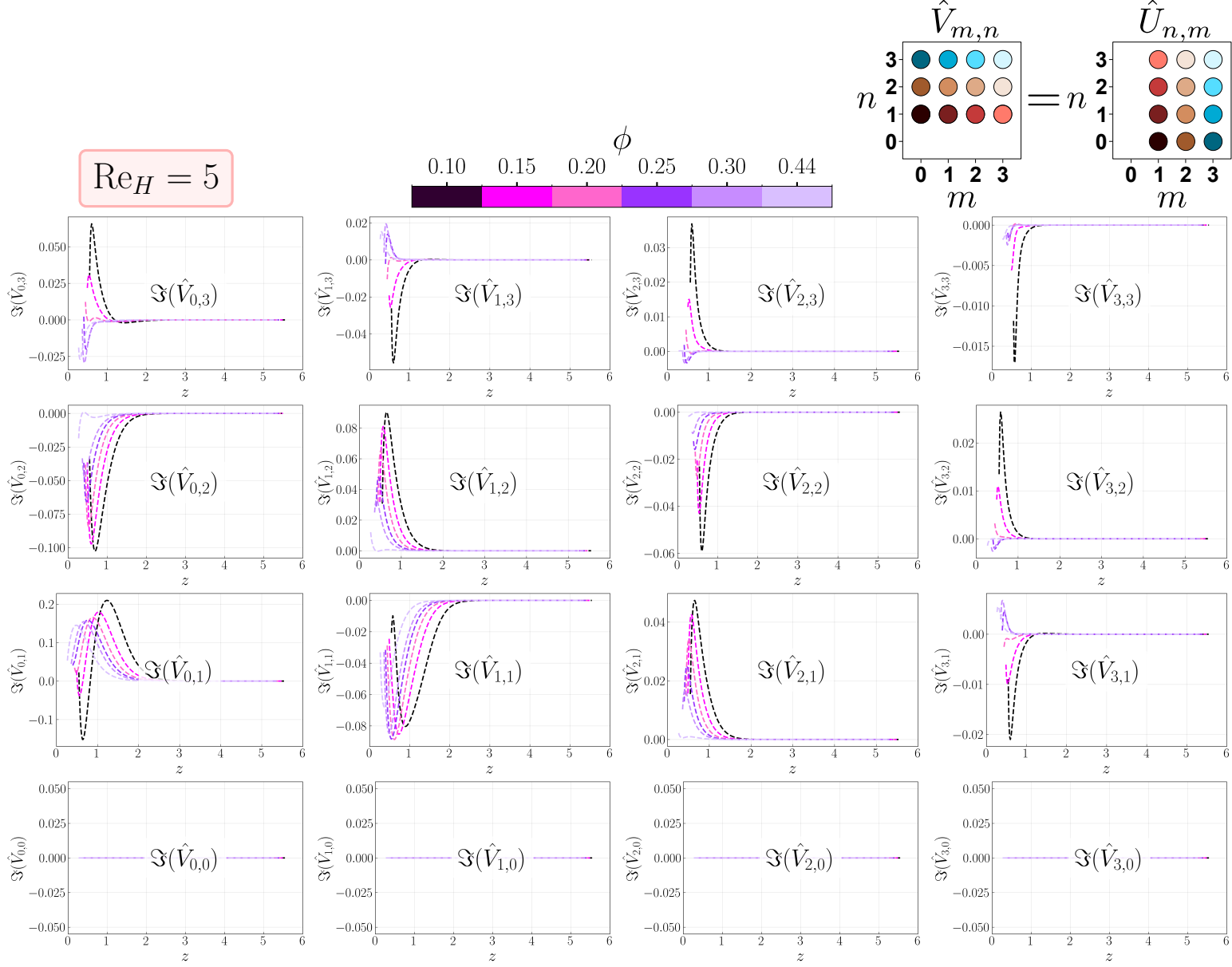


Figure B.30: Streamwise evolution of the imaginary part $\Im(\hat{V}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 5$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

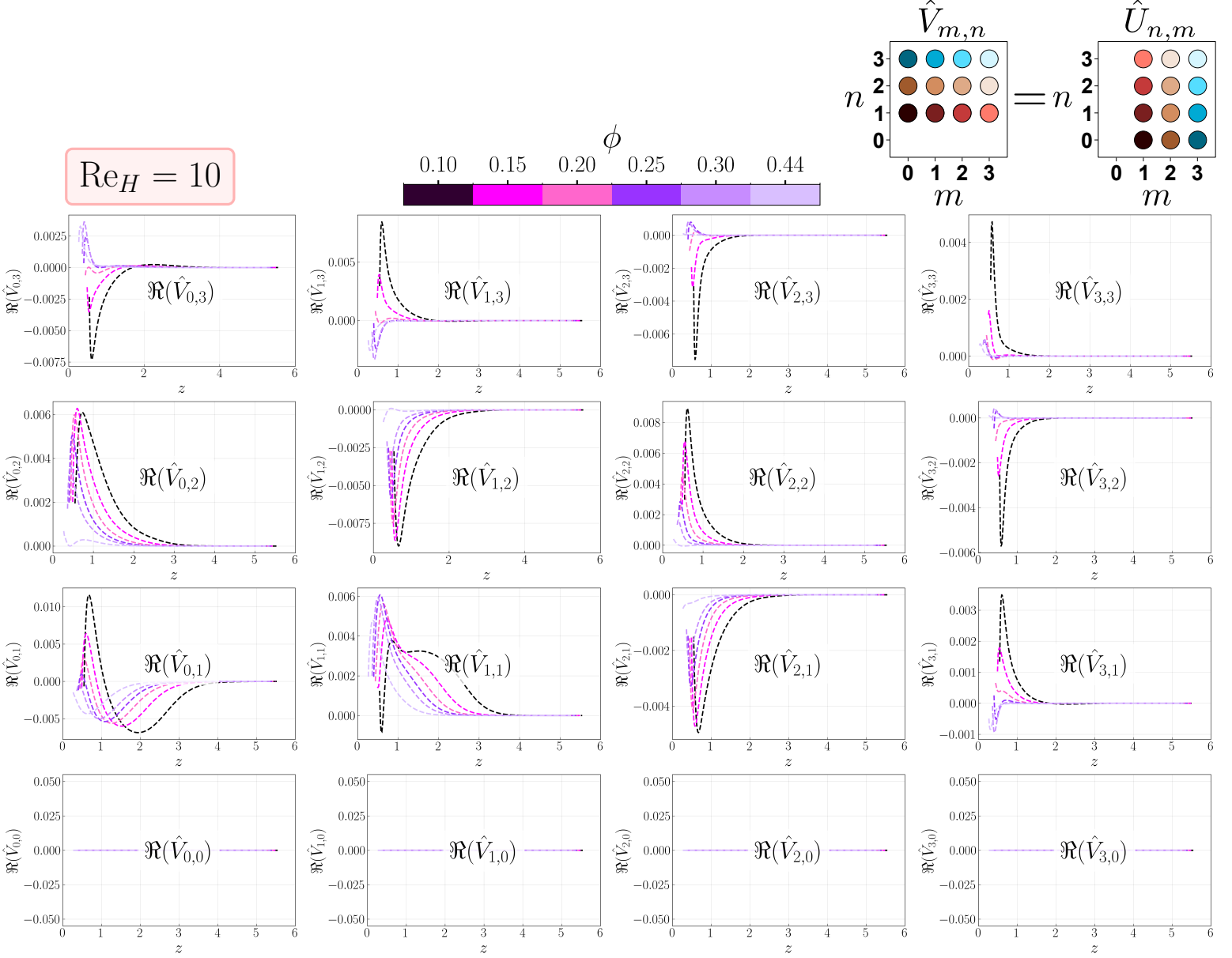


Figure B.31: Streamwise evolution of the real part $\Re(\hat{V}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 10$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

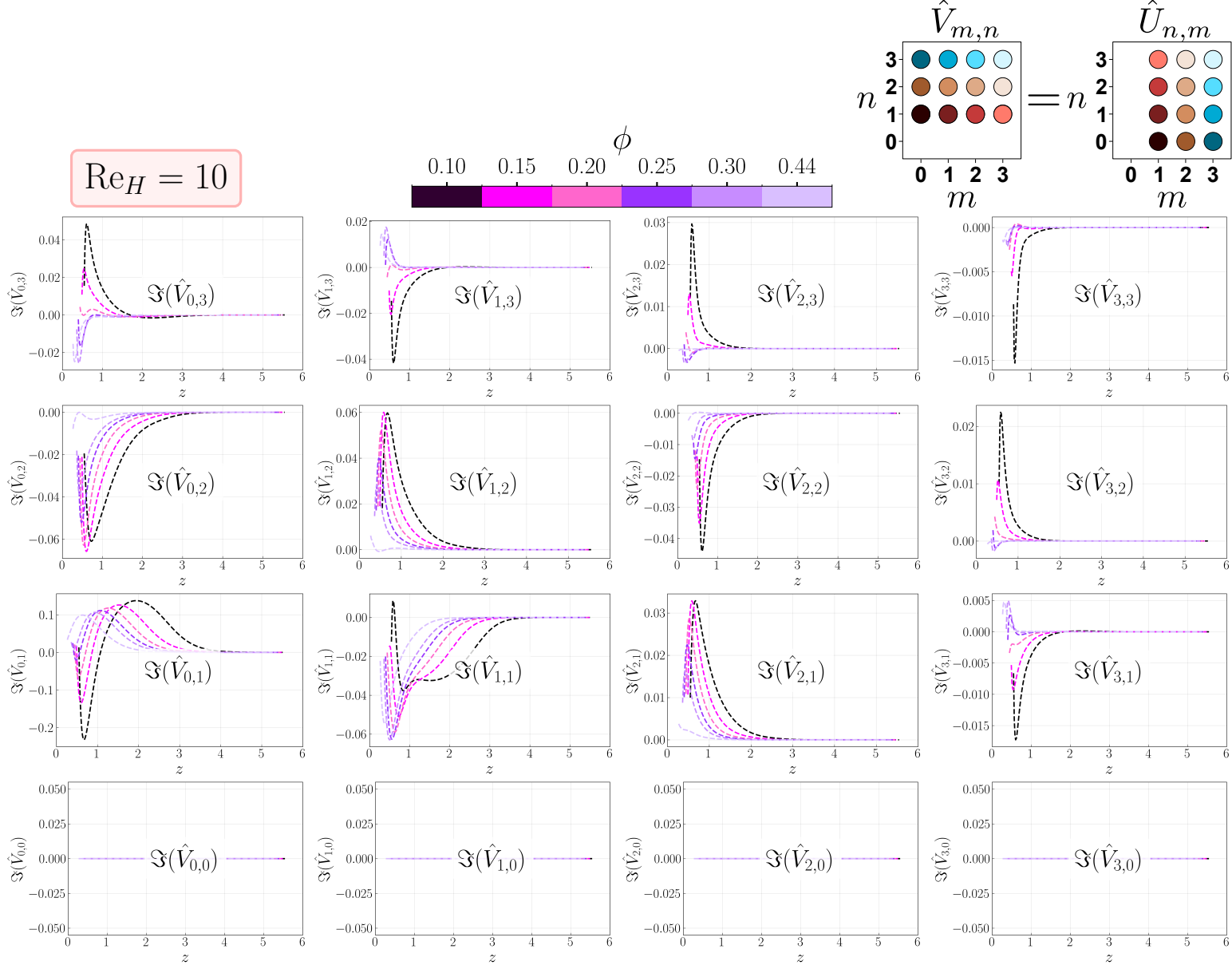


Figure B.32: Streamwise evolution of the imaginary part $\Im(\hat{V}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 10$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

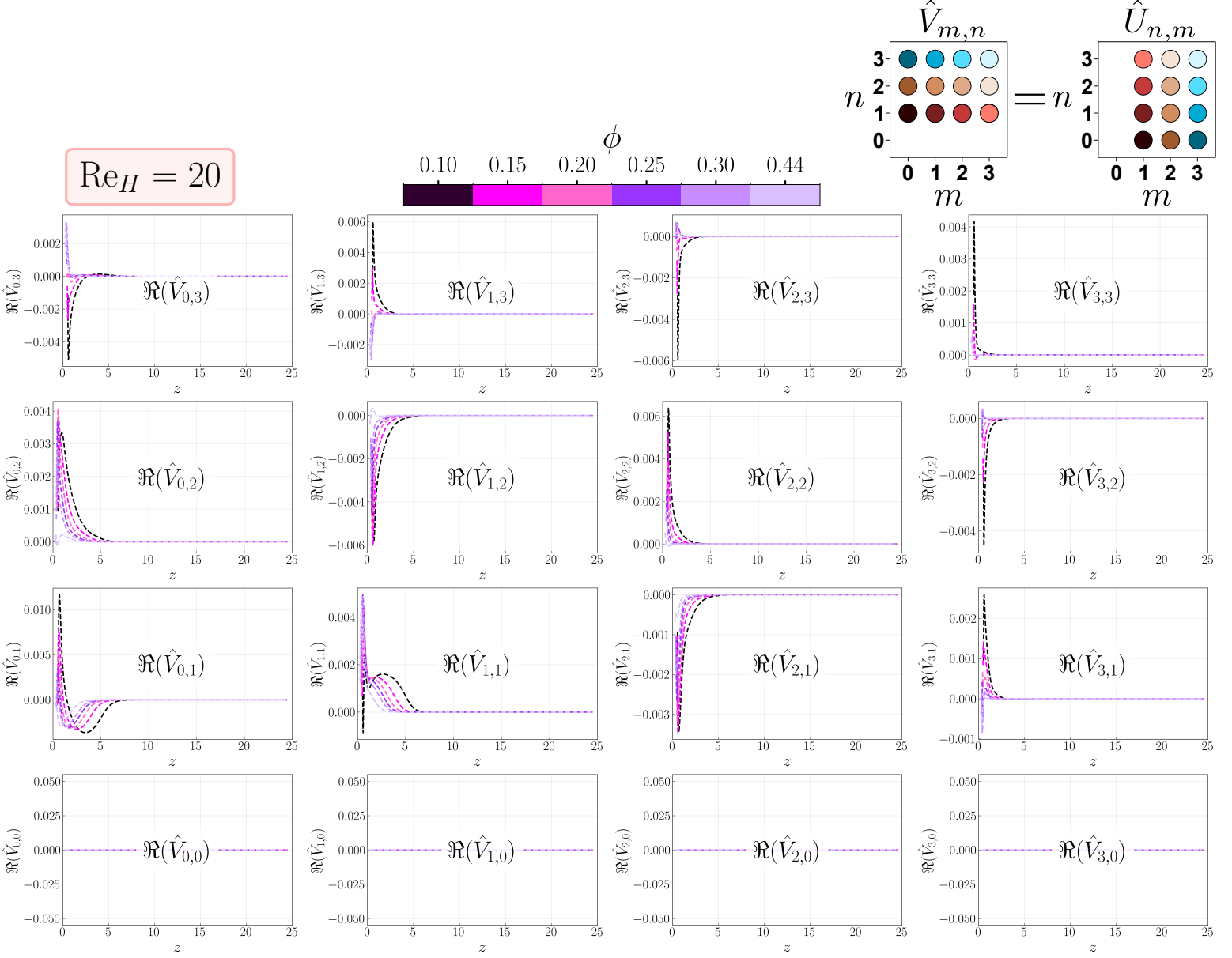


Figure B.33: Streamwise evolution of the real part $\mathcal{R}(\hat{V}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 20$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

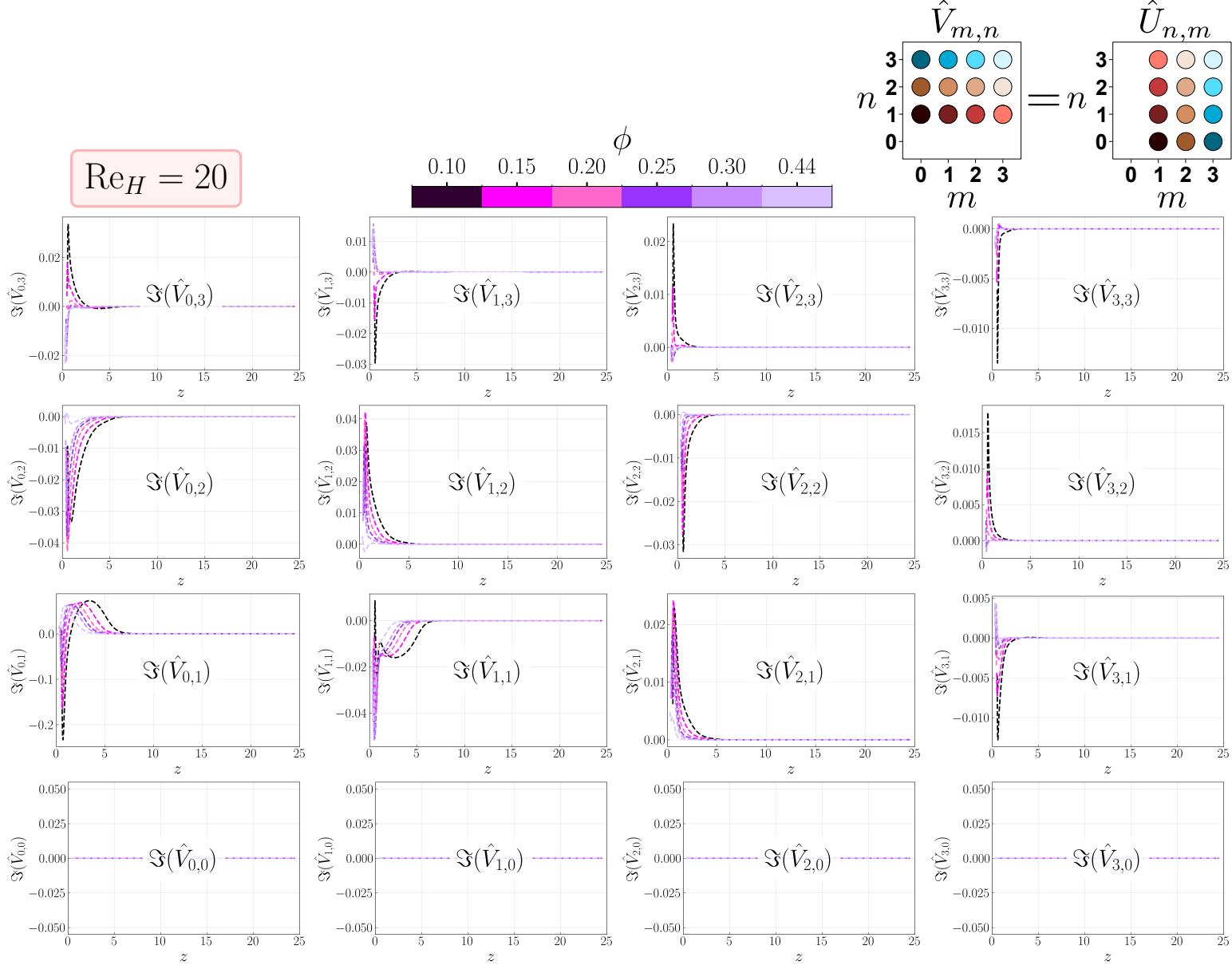


Figure B.34: Streamwise evolution of the imaginary part $\mathfrak{I}(\hat{V}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 20$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

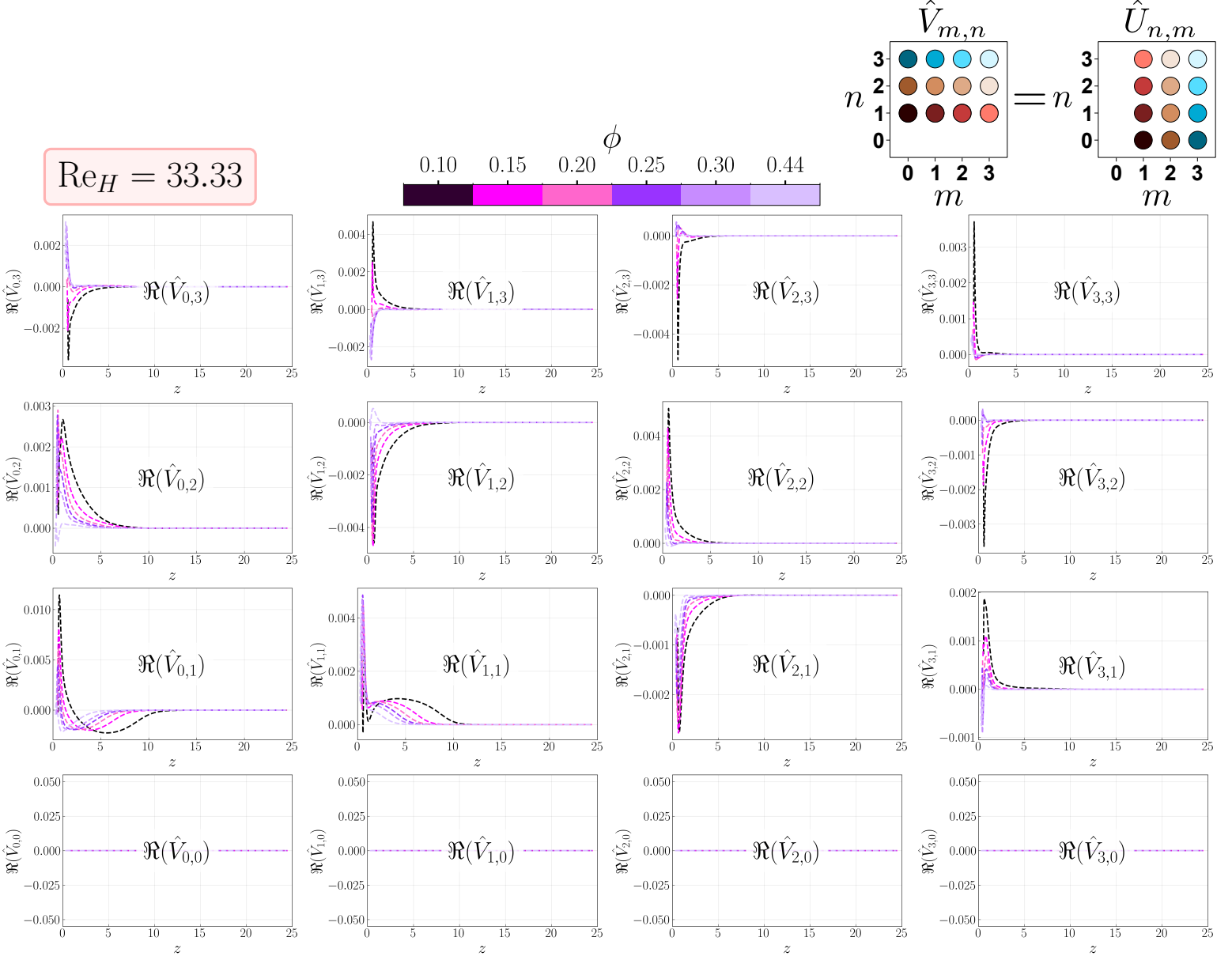


Figure B.35: Streamwise evolution of the real part $\Re(\hat{V}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 33.33$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

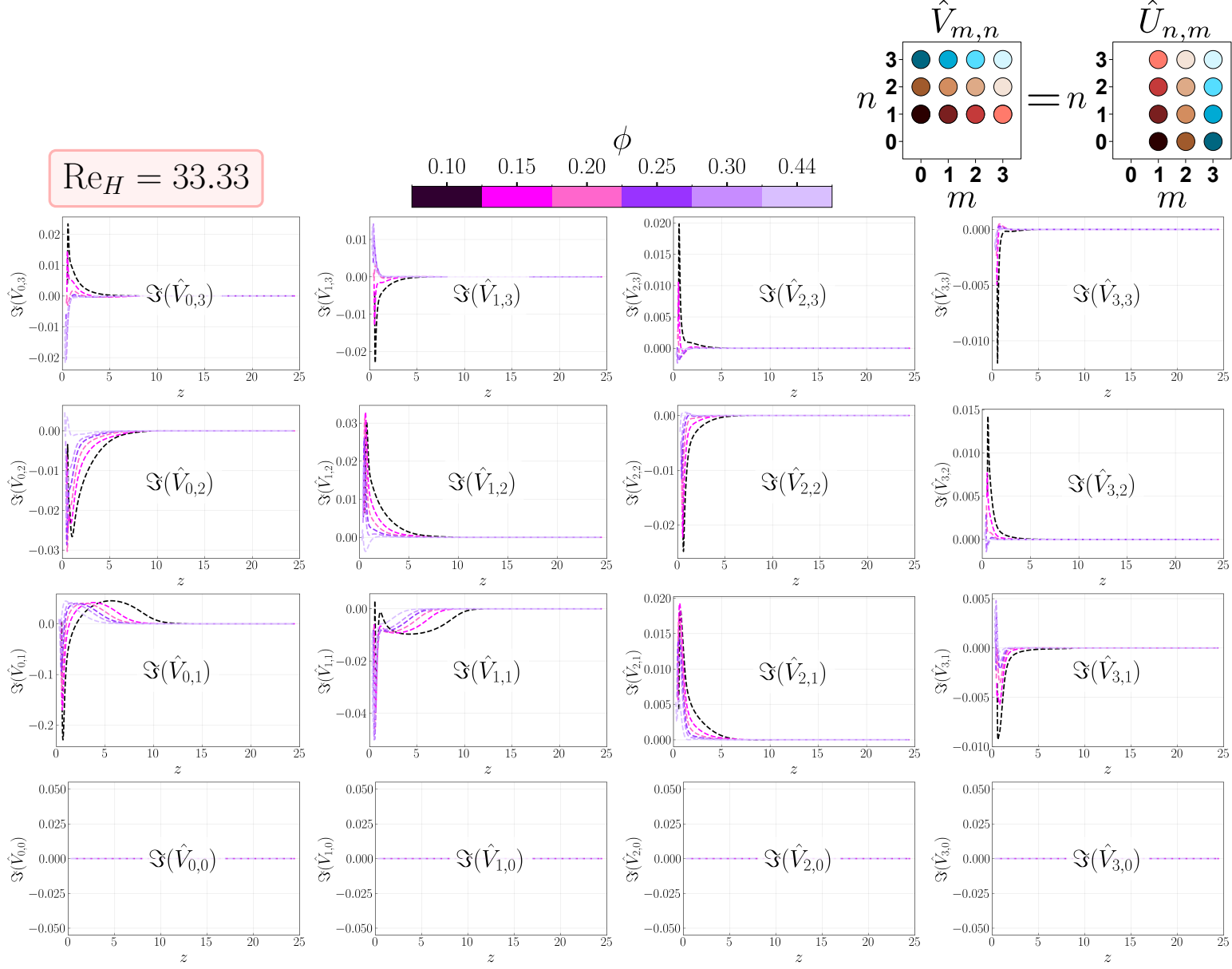


Figure B.36: Streamwise evolution of the imaginary part $\mathfrak{I}(\hat{V}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 33.33$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

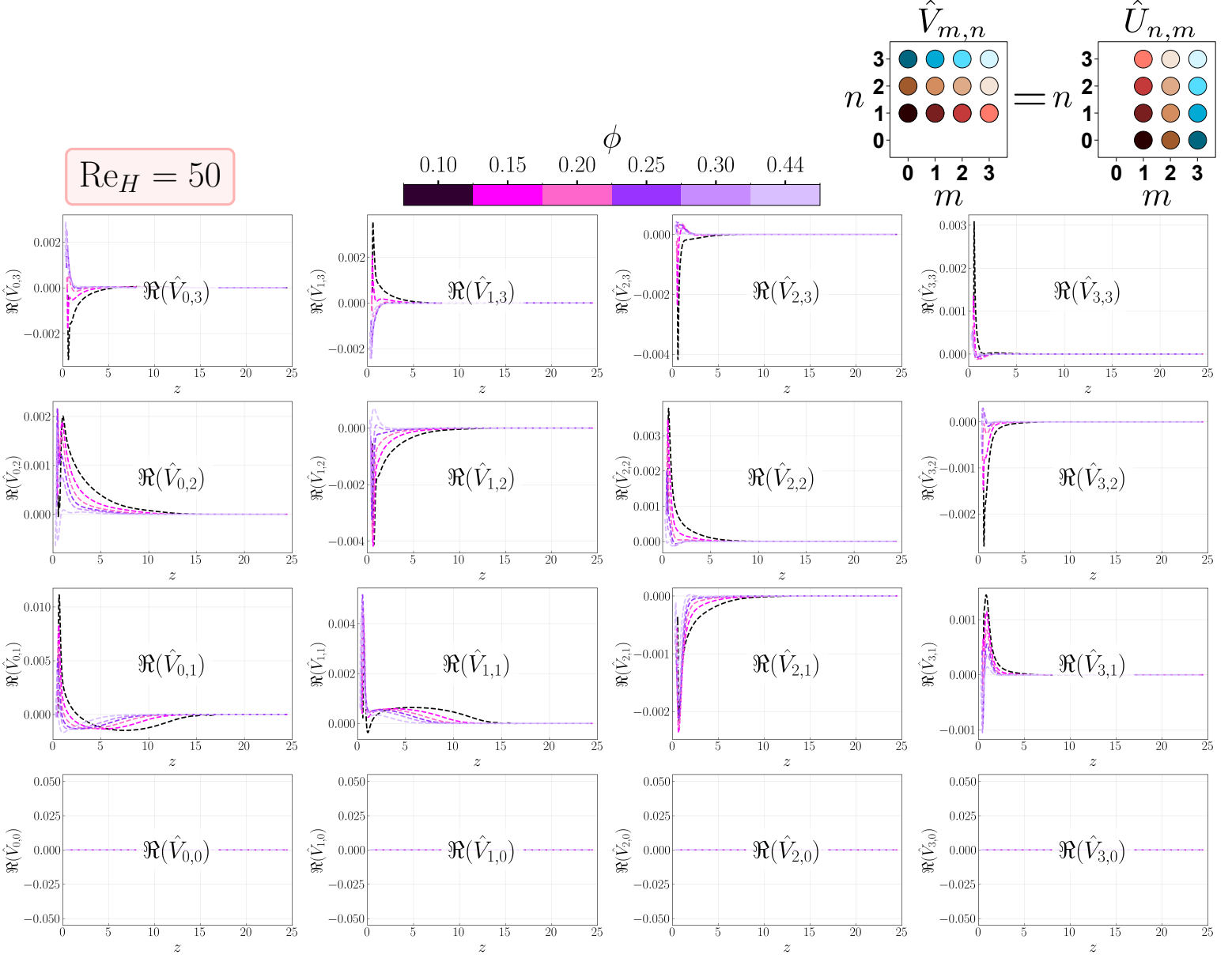


Figure B.37: Streamwise evolution of the real part $\mathcal{R}(\hat{V}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 50$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

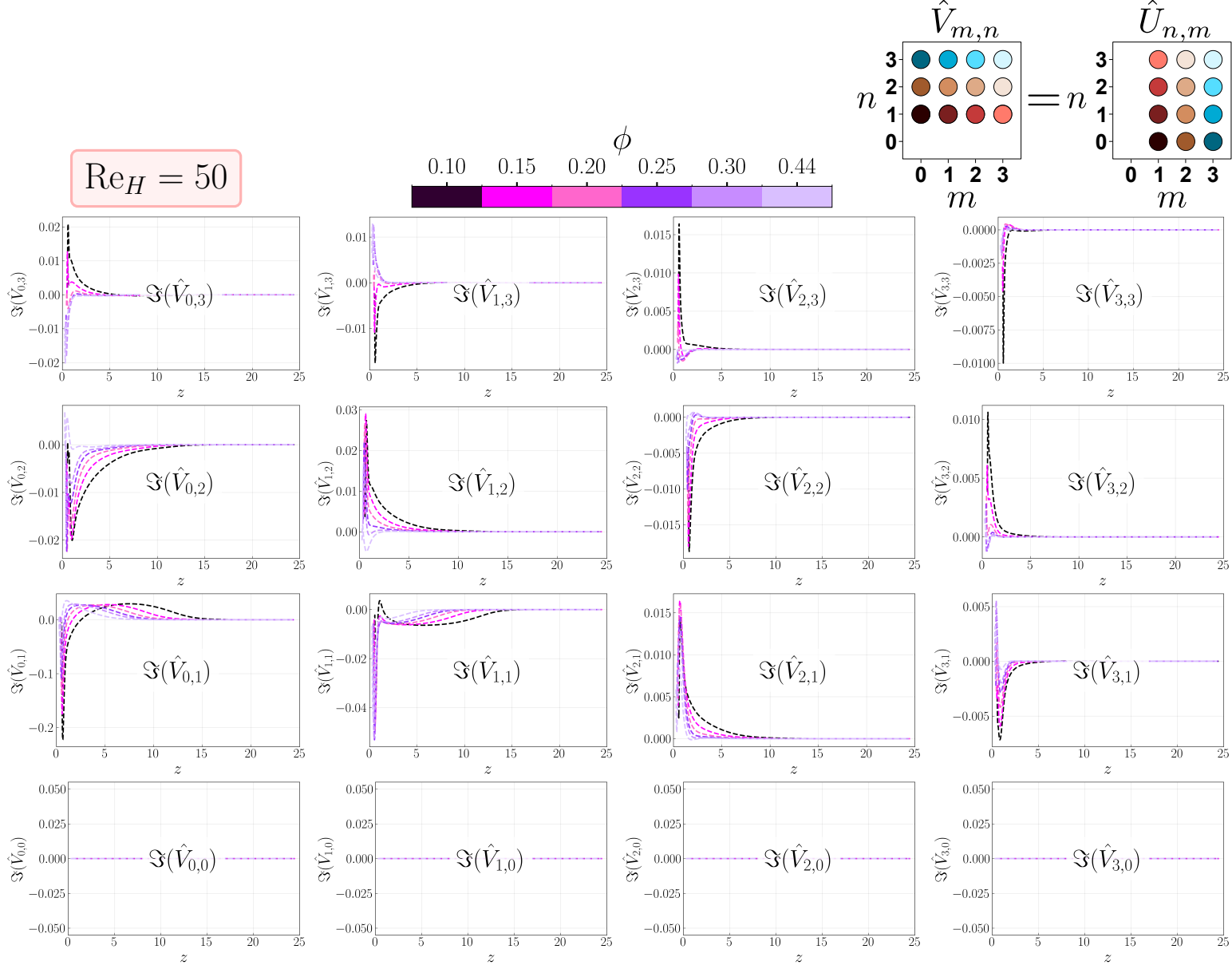


Figure B.38: Streamwise evolution of the imaginary part $\mathfrak{I}(\hat{V}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 50$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

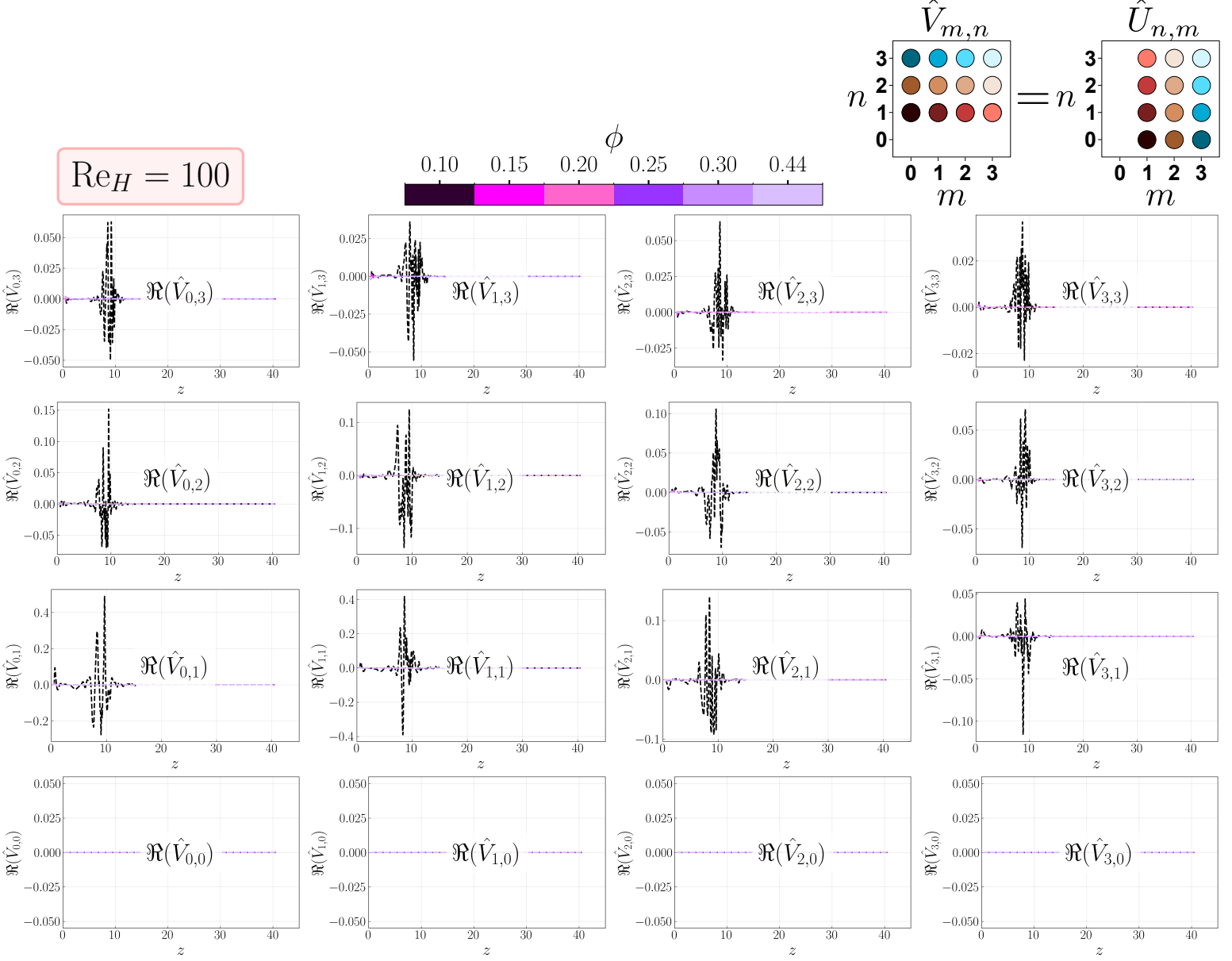


Figure B.39: Streamwise evolution of the real part $\Re(\hat{V}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 100$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the retained Fourier modes and the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$, which holds for the steady-flow cases. Deviations from this symmetry are observed for the unsteady case at $\text{Re}_H = 100$ and $\phi = 0.10$, reflecting the onset of symmetry-breaking dynamics.